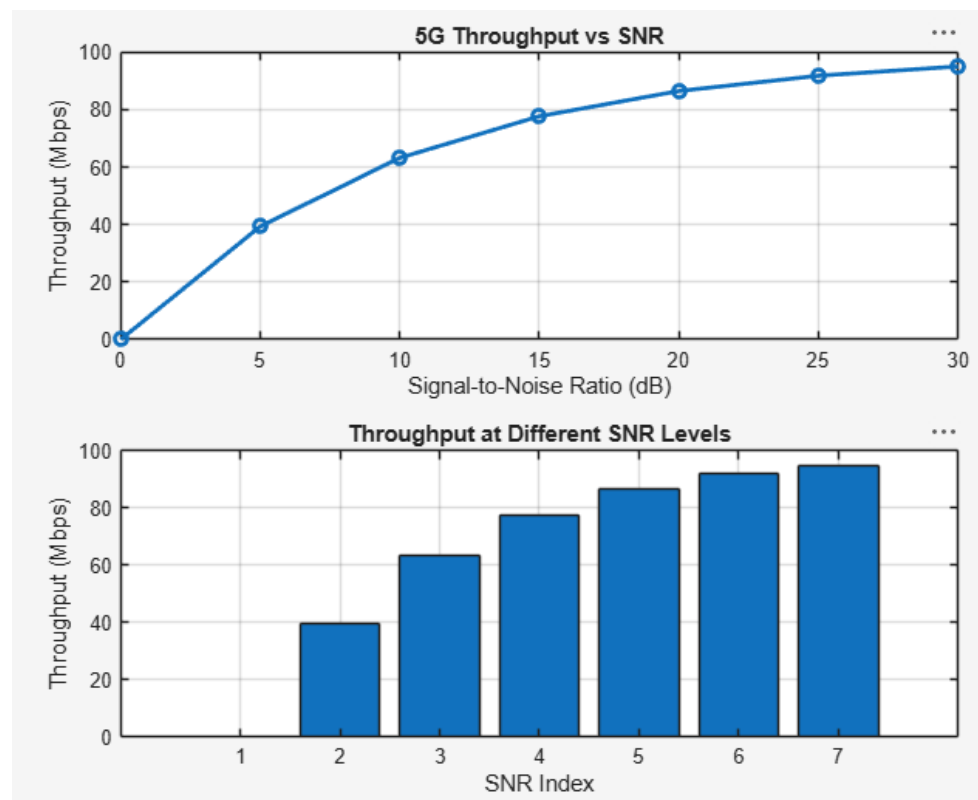


## Ex.No.14: 5G Throughput vs SNR with Modulation Labels

### Objective 1: Matlab Code and Output

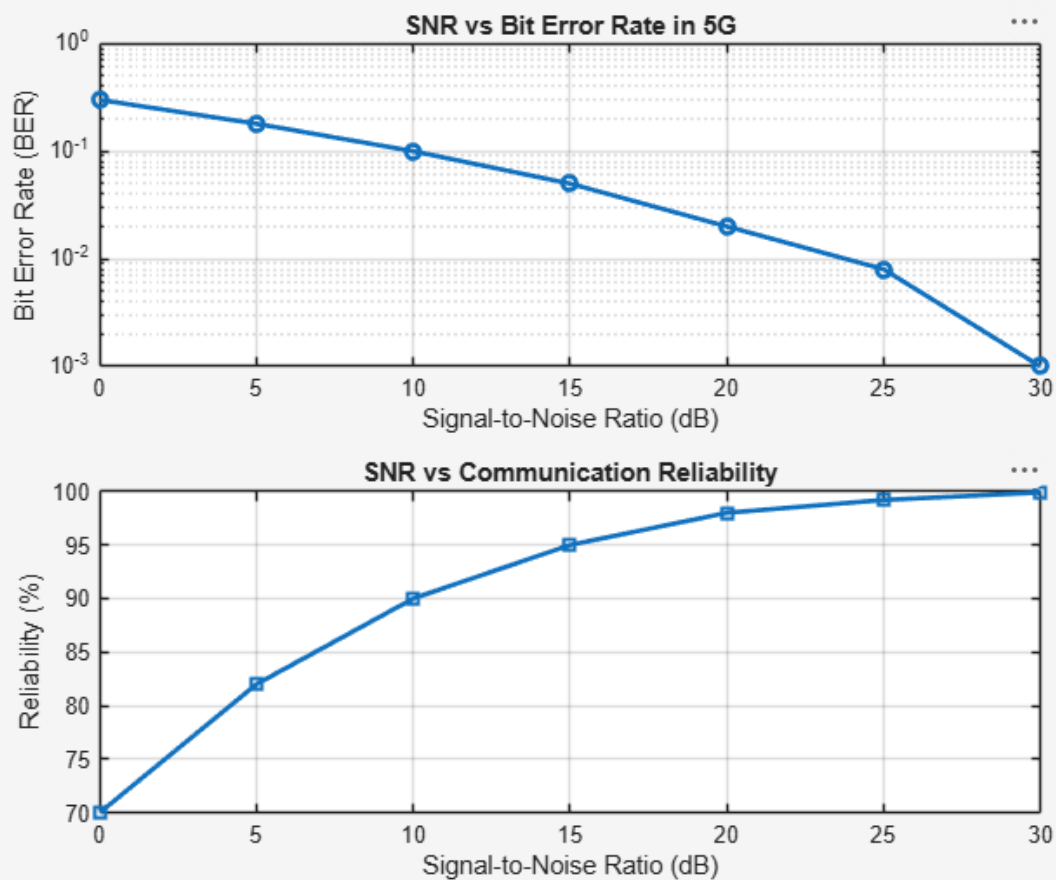
/MATLAB Drive/slotb14.m

```
1   clc;
2   clear;
3   close all;
4   % SNR values (dB)
5   SNR = 0:5:30;
6   % Maximum Throughput (Mbps)
7   Max_TP = 100;
8   % Throughput calculation
9   Throughput = Max_TP*(1-exp(-SNR/10));
10  disp('SNR(dB)    Throughput(Mbps)')
11  disp([SNR' Throughput'])
12  figure
13  % ----- Graph 1 -----
14  subplot(2,1,1)
15  plot(SNR,Throughput,'-o','LineWidth',2)
16  title('5G Throughput vs SNR')
17  xlabel('Signal-to-Noise Ratio (dB)')
18  ylabel('Throughput (Mbps)')
19  grid on
20  % ----- Graph 2 -----
21  subplot(2,1,2)
22  bar(Throughput)
23  title('Throughput at Different SNR Levels')
24  xlabel('SNR Index')
25  ylabel('Throughput (Mbps)')
26  grid on
```



## Objective 2: Matlab Code and Output

```
28 % SNR values (dB)
29 SNR = 0:5:30;
30 % Simulated Bit Error Rate (BER)
31 BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
32 % Reliability (%)
33 Reliability = (1-BER)*100;
34 disp('SNR(dB)   BER   Reliability(%)')
35 disp([SNR' BER' Reliability'])
36 figure
37 % ----- Graph 1 -----
38 subplot(2,1,1)
39 semilogy(SNR,BER,'-o','LineWidth',2)
40 title('SNR vs Bit Error Rate in 5G')
41 xlabel('Signal-to-Noise Ratio (dB)')
42 ylabel('Bit Error Rate (BER)')
43 grid on
44 % ----- Graph 2 -----
45 subplot(2,1,2)
46 plot(SNR,Reliability,'-s','LineWidth',2)
47 title('SNR vs Communication Reliability')
48 xlabel('Signal-to-Noise Ratio (dB)')
49 ylabel('Reliability (%)')
50 grid on
```



### Objective 3: Matlab Code and Output

```
52 % Modulation schemes
53 Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};
54 % Required SNR (dB)
55 SNR = [5 10 18 25];
56 % Spectral Efficiency (bits/s/Hz)
57 Efficiency = [2 4 6 8];
58 disp('Required SNR (dB) and Spectral Efficiency')
59 disp(table(Modulation,'SNR',Efficiency', ...
60     'VariableNames',{'Modulation','SNR_dB','Efficiency'}
61     figure
62     % ----- Graph 1 -----
63     subplot(2,1,2)
64     plot(SNR,Efficiency,'-o','LineWidth',2)
65     title('SNR vs Spectral Efficiency')
66     xlabel('Signal-to-Noise Ratio (dB)')
67     ylabel('Spectral Efficiency (bits/s/Hz)')
68     grid on
69     % ----- Graph 2 -----
70     subplot(2,1,1)
71     plot(1:4,SNR,'-s','LineWidth',2)
72     xticks(1:4)
73     xticklabels(Modulation)
74     title('Required SNR for Different Modulation Schemes')
75     xlabel('Modulation Scheme')
76     ylabel('Required SNR (dB)')
77     grid on
```

